Announcements

- EBIO party today
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Today

- Binomial distribution as emergent process
  - agent based model
- Using modern RNG libraries
- Events in continuous time
- Exponential distribution

Agent-based model

- Another outcome: number dead out of 100 individuals

Algorithm

```
set each individual to "alive"
set each individual to p_death = 0.2
for each individual
    generate U from U(0,1)
    if U <= p_death
        set individual to "dead"
count dead
```

Binomial distribution

- Emerges from the agent-based simulation
- Underlying process: stochastic binary events
- Number of events that occur out of N possible
- Higher-level abstraction of the biological process or ABM is possible
  - i.e. simplifies to: $N_{\mathrm{dead}} \sim \mathrm{Binomial}(N, p)$

Binomial distribution

- Sum of independent stochastic binary events with the same probability
- Pr(X = x | n, p) =

Discrete x
0,1,2...n

$$f(x,n,p) = \binom{n}{x} p^{x} (1-p)^{n-x}$$

Probability mass function (pmf)

$$X \sim \mathrm{Binomial}(n,p)$$

Binomial coefficient: how many ways to choose x from n

"Probability that x events occur out of n possible"

Modern algorithms

1. generate random bits
2. turn those into integers
3. divide by largest integer to give Uniform(0,1)

Modern algorithms

- R: Mersenne_Twister
- Numpy: PCG64 (Permuted Congruential Generator)

Discrete vs continuous time

- Discrete: all events in a period
- Continuous: each event occurs at a particular time

Events in continuous time

- When does the event occur?
  - “time to event”
  - “waiting time”

When does the individual die?

- Approximation
  - small interval of time delta t
  - constant probability of death, Pr(death)
- Diagrams
  - Event sequence diagram
  - Flowchart

When does the individual die?

Algorithm

```
set c, \delta t
set state = alive, t = 0
while state = alive
    t = t + \delta t
    generate U from U(0,1)
    if U <= p_death
        state = dead
print t, time to death
```